Montgomery projective
Tangent rule

$$X' = \frac{\left(3\left(\frac{X}{Z}\right)^2 + 2A\frac{X}{Z} + 1\right)^2}{2B\frac{Y}{Z}} \cdot B - 2\frac{X}{Z} - A$$

???

$a \cdot x^2 + y^2 = 1 + d \cdot x^2 y^2 \rightsquigarrow y^2 \cdot (dx^2 - 1) = a x^2 - 1$

Twisted edwards : $(x_1, y_1) \odot (x_2, y_2)$ $\quad y^2 = \sqrt{\frac{ax^2-1}{dx^2-1}}$

$$x' = \frac{x_1 y_2 + y_1 x_2}{1 + d x_1 x_2 y_1 y_2} \qquad y' = \frac{y_1 y_2 - a x_1 x_2}{1 - d x_1 x_2 y_1 y_2}$$

For $P = Q = (x, y)$: $P \odot Q = (x'; y')$ with:

$$x' = \frac{xy + yx}{1 + d x^2 y^2} 2xy \qquad y' = \frac{y^2 - a x^2}{1 - d x^2 y^2}$$

$$\rightsquigarrow P^3(x) = \frac{x y' + y x'}{1 + d x x' y y'} \qquad P^3(y) = \frac{y y' - a x x'}{1 - d x x' y y'}$$

$$\rightsquigarrow P^3(x) = \frac{x \cdot \left(\frac{y^2 - a x^2}{1 - d x^2 y^2}\right) + y \cdot \left(\frac{2xy}{1 + d x^2 y^2}\right)}{1 - d x \cdot \left(\frac{2xy}{1 + d x^2 y^2}\right) \cdot y \cdot \left(\frac{y^2 - a x^2}{1 - d x^2 y^2}\right)}$$

$$= \frac{\frac{x \cdot (y^2 - a x^2) \cdot (1 + d x^2 y^2) + y \cdot 2xy \cdot (1 - d x^2 y^2)}{1 - d^2 x^4 y^4}}{1 - d x \cdot \frac{x \cdot 2xy \cdot y \cdot (y^2 - a x^2)}{1 - d^2 x^4 y^4}}$$

$$= \frac{(x y^2 - a x^3) \cdot (1 + d x^2 y^2) + 2x y^2 \cdot (1 - d x^2 y^2)}{(1 - d^2 x^4 y^4) \cdot \left(1 - d x \cdot \left(\frac{2x^2 y^2 \cdot (y^2 - a x^2)}{1 - d^2 x^4 y^4}\right)\right)}$$

$$= \frac{x y^2 + d x^3 y^4 - a x^3 - a d x^5 y^2 + 2xy^2 - 2d x^3 y^4}{(1 - d^2 x^4 y^4) \cdot \left(1 - d x \cdot \left(\frac{2x^2 y^4 - 2a x^4 y^2}{1 - d^2 x^4 y^4}\right)\right)}$$

$x \cdot (3y^2 - d x^2 y^4 - a x^2 + a d x^4 y^2)$

$$= \frac{3xy^2 - d x^3 y^4 - a x^3 - a d x^5 y^2}{(\quad - y^2) \cdot (\quad - y^2)}$$

dx^3

Set $P = [X : Y : Z]$ $\qquad x(P) = [X : Z]$

Set $x\,Dbl(x(P)) = [X_{Dbl} : Z_{Dbl}]$, $\quad x(P^2) = [X' : Z']$.

To show: $x\,Dbl(x(P)) = x(P^2)$

Proof:

$X_{Dbl} = (X+Z)^2 (X-Z)^2$, $\quad X' = B(3X^2 + 2AXZ + Z^2)^2 2BYZZ^2$

$$- (AZ^2 + 2XZ)(2BYZ)^3$$

$$= B(9X^4 + 6X^2AXZ + 3X^2Z^2$$

$$+ 6X^2AXZ + 4A^2X^2Z^2 + 2AXZ^3$$

$$+ 3X^2Z^2 + 2AXZ^3 + Z^4)\,2BYZ^3$$

$$- (AZ^2\,8B^3Y^3Z^3 + 2XZ\,8B^3Y^3Z^3)$$

$$= B(9X^4 + 12AX^3Z + 6X^2Z^2 + 4AXZ^3$$

$$+ 4A^2X^2Z^2 + Z^4)\,2BYZ^3$$

$$- 8AB^3Y^3Z^5 - 16B^3XY^3Z^3$$

$$= 2B^2YZ^3 \cdot (\ldots)$$

???

Input: $m = \langle b_0, b_1, \dots, b_{\ell-1} \rangle$ where $m = \sum_{i=0}^{\ell-1} b_i 2^i$,

$P^1, P^2, P^4, P^8, \dots, P^{2^{\ell-1}}$

Output: P^m

ladder($(b_0, \dots, b_{\ell-1})$, $(P^1, \dots, P^{2^{\ell-1}})$):

$sP := O$ ← infinity ; $tP := P$

for $i = 0$ to $\ell - 1$

if $b_i = 0$ then

$sP := sP \cdot P^{2^i}$ (with xAdd (sP, P^{2^i}, tP))

P^{2^i} → "P", tP → "Q", sP → "Q − P"

Technically case $i = 0$ seperately with xDbl

else

$tP := tP \cdot P^{2^i}$ (with xAdd (tP, P^{2^i}, sP))

endif

end for

return $(sP, sP \cdot P)$

$2 \cdot 6 = 12$ constraints

+ selection ($2 \cdot 3 \cdot 1 = 6$ constraints)

⇝ 18 constr. total

(2 less than without precomp.)

TE:

$$ax^2 + y^2 = 1 + dx^2y^2$$

$$\Leftrightarrow \quad y^2 \cdot (1 - dx^2) = 1 - ax^2$$

$$\Leftrightarrow \quad y^2 = \frac{1 - ax^2}{1 - dx^2}$$

$$\Rightarrow \quad y = \pm\sqrt{\frac{1 - ax^2}{1 - dx^2}}$$

Doubling:

$$x^2 = \frac{2xy}{1 + dx^2y^2} = \frac{\pm 2x\sqrt{\frac{1-ax^2}{1-dx^2}}}{1 + dx^2 \cdot \frac{1-ax^2}{1-dx^2}} = \pm\frac{2x\sqrt{\frac{1-ax^2}{1-dx^2}} \cdot (1-dx^2)}{1 - dx^2 + dx^2 \cdot (1 - ax^2)}$$

$$= \pm\frac{2x\sqrt{(1-ax^2)\cdot(1-dx^2)}}{1 + dx^2dx^2} = \pm\frac{2x\sqrt{(1-ax^2)\cdot(1-dx^2)}}{1 + adx^4} \rightarrow *$$

$P_1 = P^2$
$P_2 = P$

$$x^3 = \pm\frac{2x\sqrt{(1-ax^2)\cdot(1-dx^2)}}{1+adx^4} \cdot \sqrt{\frac{1-ax^2}{1-dx^2}} +$$

$$* \quad y^2 = \pm\sqrt{\frac{1 - a \cdot \frac{(4x^2 \cdot (1-ax^2)\cdot(1-dx^2)}{(1+adx^4)^2}}{1 - d \cdot \frac{(4x^2 \cdot (1-ax^2)\cdot(1-dx^2)}{(1+adx^4)^2}}} = *$$ see next page

$$y^2 = \frac{y^2 - ax^2}{1 - dx^2y^2} = \frac{\pm\sqrt{\frac{1-ax^2}{1-dx^2}} - dx^2}{1 - dx^2 \cdot \frac{1-ax^2}{1-dx^2}} = \frac{\left(\pm\sqrt{\frac{1-ax^2}{1-dx^2}} - ax^2\right)(1-dx^2)}{1 - dx^2 - dx^2 \cdot (1 - ax^2)}$$

$$= \frac{\pm\sqrt{(1-ax^2)\cdot(1-dx^2)} - ax^2\cdot(1-dx^2)}{1-dx^2\cdot(1+1-ax^2)}$$

$$= \frac{\pm\sqrt{(1-ax^2)(1-dx^2)} - ax^2(1-dx^2)}{1-2dx^2+adx^4}$$

$$* \quad y^2 = \sqrt{\frac{(1+adx^4)^2 - 4ax^2(1-ax^2)\cdot(1-dx^2)}{(1+adx^4)^2 - 4dx^2\cdot(1-ax^2)\cdot(1-dx^2)}}$$

$$(1-ax^2)\cdot(1-dx^2)$$
$$= 1-ax^2-dx^2+adx^4$$